

Email Phishing Detection Using Machine Learning: A Comparison of Naive Bayes, Logistic Regression, and Support Vector Machines

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Abstract—Phishing emails remain one of the most common vectors for fraud and data theft, and their growing sophistication allows many to slip past traditional rule-based filters. This work frames phishing detection as a supervised text-classification problem and compares three classical machine-learning classifiers—Multinomial Naive Bayes, Logistic Regression, and a linear Support Vector Machine (SVM)—on the CEAS_08 subset of a public phishing email corpus. Email subject and body text are combined and transformed into features using term frequency-inverse document frequency (TF-IDF), and the models are evaluated on accuracy, precision, recall, F_1 -score, and ROC-AUC using a stratified train/test split. All three models exceed 98% F_1 , with the linear SVM performing best ($F_1 = 0.997$, ROC-AUC = 0.9998). An exploratory analysis shows that message length is a far stronger discriminator than the number of embedded URLs, which is nearly identical across classes in this corpus. We also discuss a limitation: because the legitimate emails are dominated by technical mailing-list traffic, the near-perfect scores may not fully transfer to a general inbox.

Index Terms—phishing detection, email classification, machine learning, TF-IDF, support vector machine, naive Bayes, logistic regression

I. INTRODUCTION

Email continues to be one of the most heavily exploited communication channels for phishing, scams, and unsolicited spam, imposing significant financial and security costs on both individuals and organizations. Attackers craft messages that impersonate trusted senders and pressure recipients into revealing credentials or transferring funds. As these messages grow more sophisticated, static rule-based filters—which rely on fixed keyword lists or blocklists—become increasingly unreliable and difficult to scale.

Machine learning offers an attractive alternative: rather than encoding rules by hand, a classifier can learn the statistical patterns that distinguish phishing from legitimate email directly from labeled data. This paper investigates that approach using three widely used classical classifiers. Our contributions are as follows. First, we build a reproducible preprocessing and feature-extraction pipeline based on TF-IDF. Second, we train and evaluate Naive Bayes, Logistic Regression, and a linear SVM under an identical experimental protocol so that

they can be compared on equal footing. Third, we report an exploratory data analysis that identifies which raw email properties actually carry signal, and we discuss the practical limitations of the resulting models.

II. RELATED WORK

Early email filters relied largely on handcrafted rules and blocklists, which are brittle against novel attacks. Statistical text classification later became the dominant approach. Naive Bayes was among the first probabilistic methods applied to spam and text categorization [1], while support vector machines were shown to be highly effective for high-dimensional sparse text representations [2]. Surveys of phishing-specific filtering [3] catalog a broad range of content-based and behavioral features. Our study follows this established line of work, pairing the classic TF-IDF representation [4] with three standard classifiers and focusing on a controlled, reproducible comparison rather than on novel model architectures.

III. DATASET

We use the CEAS_08 subset of a publicly available phishing email dataset hosted on Kaggle [7], which aggregates several well-known corpora (Enron, Ling, CEAS, Nazario, Nigerian Fraud, and SpamAssassin). Each record contains the sender, receiver, date, subject, body, a count of embedded URLs, and a binary label indicating whether the message is phishing or legitimate.

Before modeling, we cleaned the data by filling missing subject and body fields prior to concatenation—an important step, since naively joining a present subject with a missing body produces a null value and discards the subject entirely. We then stripped whitespace, removed empty records, and dropped exact-duplicate messages. After cleaning, the dataset contains 39,115 emails. Table I summarizes its composition, and Fig. 1 shows the class distribution: the corpus is moderately balanced, with a slight majority of phishing messages.

A. Exploratory Data Analysis

We examined two intuitive raw features before modeling. The first is the number of embedded URLs per message.

TABLE I
DATASET COMPOSITION AFTER CLEANING

Class	Count	Proportion
Legitimate	17,309	44.3%
Phishing	21,806	55.7%
Total	39,115	100%

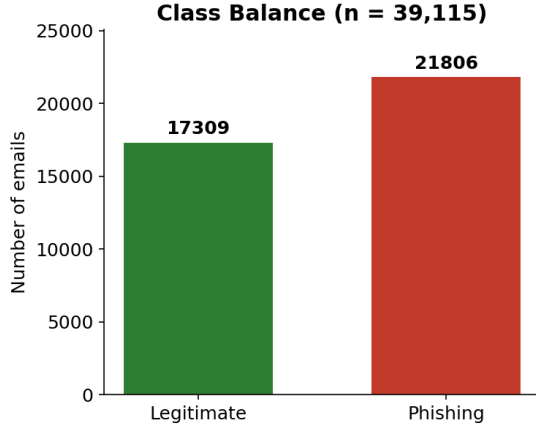


Fig. 1. Class distribution of the cleaned CEAS_08 dataset. Phishing emails form a slight majority (55.7%).

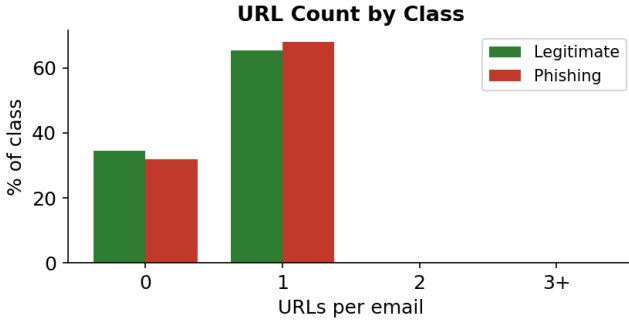


Fig. 2. Distribution of embedded URL counts by class. The two classes are nearly indistinguishable on this feature.

Contrary to the common assumption that phishing emails contain more links, Fig. 2 shows that URL counts are almost identical across the two classes (mean of 0.66 for legitimate versus 0.68 for phishing), indicating that URL volume alone is a weak signal in this corpus.

The second feature is message length. As shown in Fig. 3, phishing emails are markedly shorter than legitimate ones: the median phishing message contains 48 words compared with 190 words for legitimate mail. This large separation suggests that the textual content itself, which TF-IDF captures directly, is the more promising basis for classification.

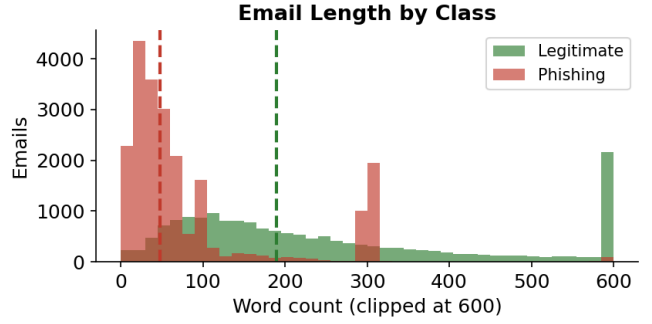


Fig. 3. Distribution of message length (word count) by class. Dashed lines mark the per-class medians; phishing emails are substantially shorter.

IV. METHODOLOGY

A. Feature Extraction

For each email we concatenate the subject and body into a single text field and convert it into a numerical feature vector using TF-IDF with English stop-word removal and a vocabulary capped at 5,000 terms. TF-IDF weights each term by how often it appears in a message, discounted by how common it is across the corpus, producing a sparse, non-negative representation that is well suited to the classifiers used here.

B. Classification Models

We evaluate three classical models. *Multinomial Naive Bayes* is a probabilistic classifier that selects the class c maximizing $P(c) \prod_i P(t_i | c)$ over the terms t_i ; the multinomial variant is designed for the count-like, non-negative structure of TF-IDF vectors. *Logistic Regression* is a linear model that estimates a phishing probability $\hat{p} = \sigma(\mathbf{w}^\top \mathbf{x} + b)$, where $\sigma(z) = 1/(1 + e^{-z})$; it is fast on high-dimensional text and interpretable, since each learned weight indicates how strongly a term pushes toward the phishing class. The *linear Support Vector Machine* [5] seeks the maximum-margin hyperplane separating the two classes in the high-dimensional feature space; a linear kernel is used because kernelized SVMs scale poorly on large, sparse text data whereas the linear formulation is efficient for exactly this case.

C. Evaluation Protocol

All models are trained and evaluated on an identical stratified 80/20 train/test split (7,823 test emails), preserving the class ratio in both partitions. We report accuracy, precision, recall, F_1 -score, and ROC-AUC. Recall is emphasized alongside precision because a missed phishing email (a false negative) is typically the more costly error in an email security setting.

V. EXPERIMENTS AND RESULTS

Table II reports the performance of all three models on the held-out test set. Every model exceeds 98% F_1 , confirming that TF-IDF features are highly informative for this task. The linear SVM achieves the best results on four of the five

TABLE II
MODEL COMPARISON ON THE TEST SET (BEST VALUE PER METRIC IN BOLD)

Model	Acc.	Prec.	Rec.	F_1	AUC
Logistic Regression	0.9942	0.9947	0.9950	0.9948	0.9996
Naive Bayes	0.9821	0.9967	0.9711	0.9837	0.9990
Linear SVM	0.9967	0.9972	0.9968	0.9970	0.9998

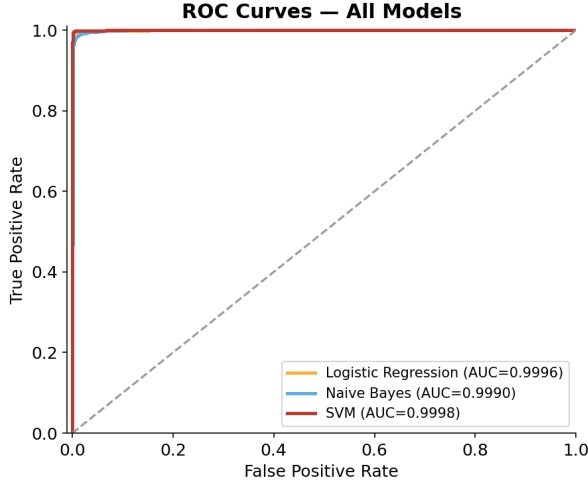


Fig. 4. ROC curves for all three models on the test set. The SVM and Logistic Regression curves sit highest, both reaching an AUC of roughly 0.9998 and 0.9996 respectively.

metrics, including the highest F_1 -score (0.997) and ROC-AUC (0.9998). Naive Bayes attains the highest precision (0.997) but the lowest recall (0.971), meaning it rarely mislabels a legitimate email yet allows comparatively more phishing to pass. Logistic Regression is a close, well-balanced second across every metric.

The ROC curves in Fig. 4 make the ranking visually clear: the SVM and Logistic Regression curves rise almost immediately to the top-left corner, indicating near-perfect separation between phishing and legitimate emails across all decision thresholds. The per-class error structure is shown in Fig. 5. Consistent with Table II, the SVM produces the fewest total misclassifications, while the Naive Bayes matrix shows a visibly larger false-negative count (phishing predicted as legitimate), reflecting its lower recall.

Finally, for all three models the training and test accuracies are nearly identical (for example, 0.9996 versus 0.9967 for the SVM), which indicates that the classifiers generalize well to unseen data within this dataset rather than overfitting the training partition.

VI. DISCUSSION

The experiments support three observations. First, the choice of representation matters more than the choice among these classifiers: once emails are encoded with TF-IDF, all three models perform strongly, and the differences between

them are small. Second, the exploratory analysis in Fig. 2 and Fig. 3 shows that not all intuitive features are useful—URL count barely separates the classes here, whereas message length separates them sharply. This is a useful reminder to verify feature assumptions against the data rather than trusting them a priori.

Third, and most important for interpreting the near-perfect scores, the legitimate portion of CEAS_08 is dominated by technical mailing-list traffic (for instance, developer and open-source discussion lists). As a result, part of what the classifiers learn is the distinction between mailing-list style and everything else, rather than a fully general notion of “legitimate” email. The reported metrics should therefore be read as an upper bound that may not transfer directly to a heterogeneous corporate or personal inbox. We also observed that adding simple engineered numeric features on top of TF-IDF produced negligible improvement, because the text representation already captures the dominant signal.

VII. CONCLUSION AND FUTURE WORK

We presented a controlled comparison of Naive Bayes, Logistic Regression, and a linear SVM for phishing email detection using TF-IDF features on the CEAS_08 dataset. All three models achieved above 98% F_1 , with the linear SVM performing best overall. Our analysis highlighted message length as a strong discriminator, URL count as a weak one, and the dataset’s mailing-list composition as a key limitation on generalization.

Future work will address that limitation and extend the approach in several directions: incorporating sender-domain reputation and URL-structure features, replacing TF-IDF with transformer-based text embeddings, training on a larger and more heterogeneous combination of corpora to improve generalization, and investigating deployment as a real-time email-filtering system.

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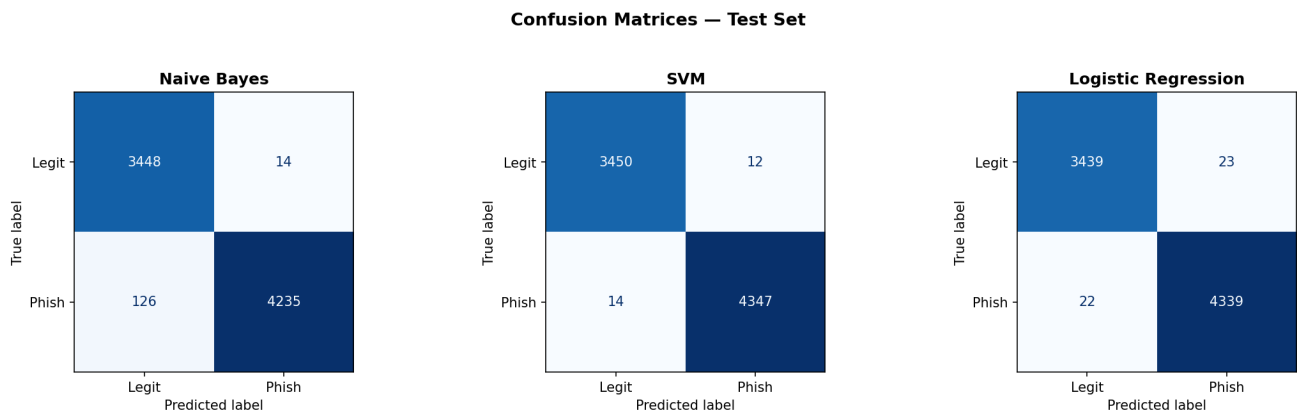


Fig. 5. Confusion matrices on the test set for Naive Bayes, the linear SVM, and Logistic Regression. The SVM yields the fewest misclassifications; Naive Bayes shows the largest number of missed phishing emails (false negatives).